MATH 101 Final Exam

Practice 1 | Version 202610122020.46

Name: _____ Date: _____

Complete and show all work on separate paper. Write exact final answers at right. Basic four-function calculator only.

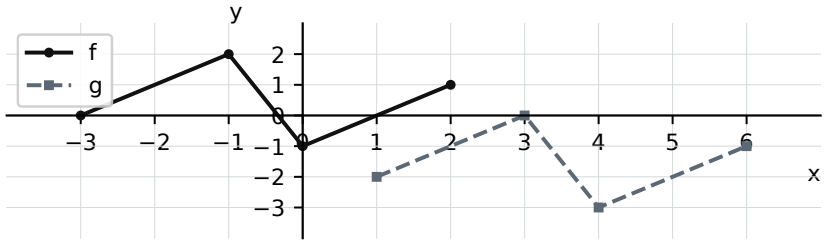
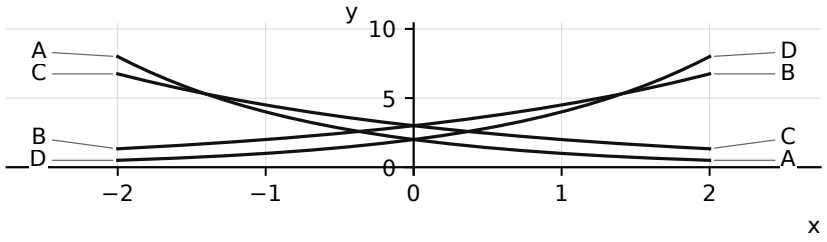
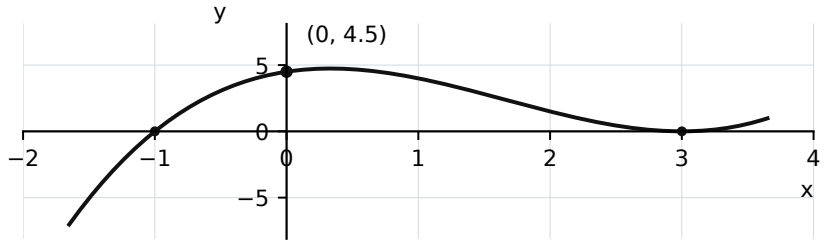
QUESTIONS	ANSWERS
1. Solve the nonlinear system algebraically. $x^2 + y^2 = 5$ $y = x + 1$	1. Intersections 10 pts
2. Which pair satisfies h(x) = f(g(x))? $h(x) = (3x + 3)^2 + 4$ $A.\quad f(x) = x^2 + 4,\ g(x) = 3x + 3$ $B.\quad f(x) = x^2 + 3,\ g(x) = 3x + 4$ $C.\quad f(x) = 3x^2 + 4,\ g(x) = x + 3$ $D.\quad f(x) = 3x + 3,\ g(x) = x^2 + 4$	2. Choice 10 pts
3. Evaluate exactly using logarithmic properties; show the property used. Here log means base 10. $(a)\quad \log(0.25) + \log(400)$ $(b)\quad \frac{\log_6(27)}{\log_6(9)}$	3. (a) 5 pts (b) 5 pts
4. Find (a) domain and range; (b) vertical asymptote and x-intercept; (c) behavior at both ends of the domain. $f(x) = \log_2(x - 2) + 1$	4. (a) Domain 2 pts (a) Range 2 pts (b) Asymptote 1 pts (b) x-intercept 1 pts (c) Left end 2 pts (c) Right end 2 pts

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QUESTIONS	ANSWERS
5. The solid graph is f and the dashed graph is g. Select the expression for g(x). $A.\quad g(x) = f(x+4) + 2$ $B.\quad g(x) = f(x-4) - 2$ $C.\quad g(x) = f(x-4) + 2$ $D.\quad g(x) = f(x+4) - 2$ 	5. Choice 10 pts
6. Match each function to graph A, B, C, or D. Each letter is used once. $(a)\quad 3(\frac{2}{3})^x$ $(b)\quad 2(\frac{1}{2})^x$ $(c)\quad 2(2)^x$ $(d)\quad 3(\frac{3}{2})^x$ 	6. (a) 3 pts (b) 2 pts (c) 3 pts (d) 2 pts
7. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form. 	7. Polynomial 10 pts

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QUESTIONS	ANSWERS
8. Find (a) domain and range; (b) both intercepts; (c) behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. $f(x) = \sqrt{x+4} - 2$	8. (a) Domain 2 pts (a) Range 2 pts (b) x-intercept 1 pts (b) y-intercept 1 pts (c) Infinite end 3 pts (c) Finite end 1 pts
9. Find all zeros: (a) list Rational Root Theorem candidates, then test only 3 and -3; (b) use polynomial long division by the confirmed linear factor; (c) use the quadratic formula on the quotient. Record all zeros. $p(x) = x^3 - 2x^2 - 4x + 3$	9. (a) Candidates/root 2 pts (b) Quotient 4 pts (c) Other zeros 4 pts
10. Find (a) domain; (b) hole, vertical asymptote, and horizontal asymptote; (c) both intercepts. $f(x) = \frac{(x+2)(x+3)}{(x-5)(x+2)}$	10. (a) Domain 2 pts (b) Hole 2 pts (b) Vertical 1 pts (b) Horizontal 1 pts (c) x-intercept 2 pts (c) y-intercept 2 pts

Instructor answer key: MATH 101 Final Exam

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Each column matches the vertical answer positions on its corresponding test page.

Concise grading reference; required work and checks are assessed on separate paper.

TEST PAGE 1

1. Intersections

$(1,\ 2);\ (-2,\ -1)$

10 pts

2. Choice

A

10 pts

3. (a)

2

5 pts

(b)

$\frac{3}{2}$

5 pts

4. (a) Domain

$(2, \infty)$

2 pts

(a) Range

$(-\infty, \infty)$

2 pts

(b) Asymptote

$x = 2$

1 pts

(b) x-intercept

$(\frac{5}{2},\ 0)$

1 pts

(c) Left end

$x \to 2^+ :\ f(x) \to -\infty$

2 pts

(c) Right end

$x \to \infty :\ f(x) \to \infty$

2 pts

TEST PAGE 2

5. Choice

B

10 pts

6. (a)

C

3 pts

(b)

A

2 pts

(c)

D

3 pts

(d)

B

2 pts

7. Polynomial

$\frac{1}{2}(x+1)(x-3)^2$

10 pts

TEST PAGE 3

8. (a) Domain

$[-4, \infty)$

2 pts

(a) Range

$[-2, \infty)$

2 pts

(b) x-intercept

$(0,\ 0)$

1 pts

(b) y-intercept

$(0,\ 0)$

1 pts

(c) Infinite end

$x \to \infty :\ f(x) \to \infty$

3 pts

(c) Finite end

$x \to -4^+ :\ f(x) \to -2$

1 pts

9. (a) Candidates/root

$\{-3, -1, 1, 3\};\ x = 3$

2 pts

(b) Quotient

$x^2 + x - 1;\ R = 0$

4 pts

(c) Other zeros

$x = -\frac{1}{2} + \frac{\sqrt{5}}{2},\ -\frac{\sqrt{5}}{2} - \frac{1}{2}$

4 pts

10. (a) Domain

$\mathbb{R} \setminus \{-2, 5\}$

2 pts

(b) Hole

$(-2,\ -\frac{1}{7})$

2 pts

(b) Vertical

$x = 5$

1 pts

(b) Horizontal

$y = 1$

1 pts

(c) x-intercept

$(-3,\ 0)$

2 pts

(c) y-intercept

$(0,\ -\frac{3}{5})$

2 pts